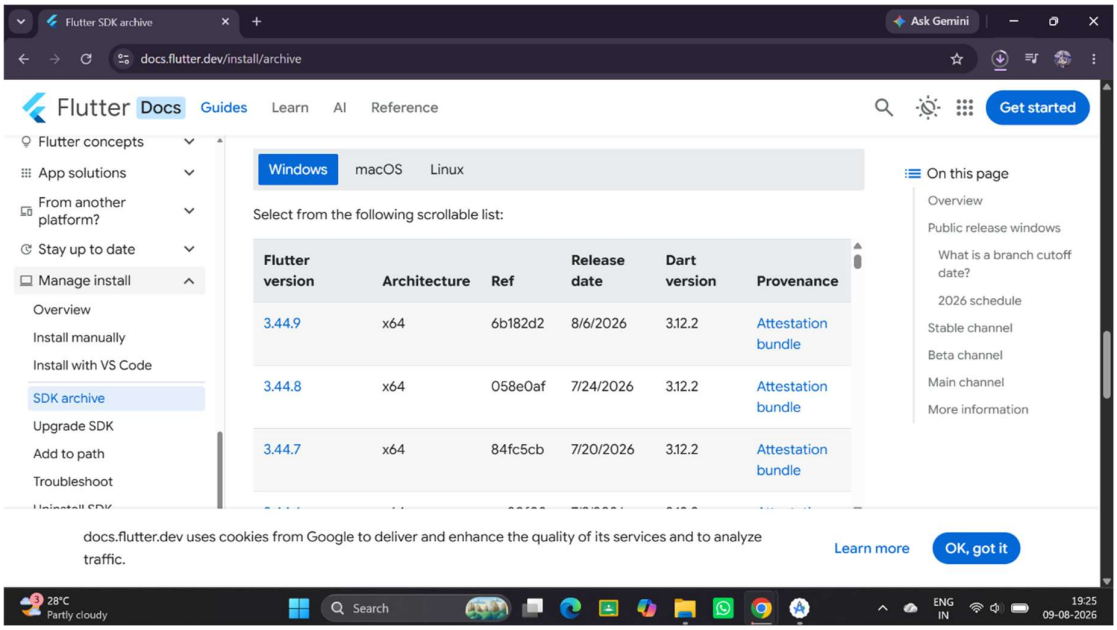
1a). To install Flutter SDK and Dart SDK, and configure the development environment.

Procedure:

Step 1: Download Flutter SDK & Dart SDK

- Visit the official Flutter website.
- Download the latest Flutter SDK.
- Save the ZIP file to the local system.

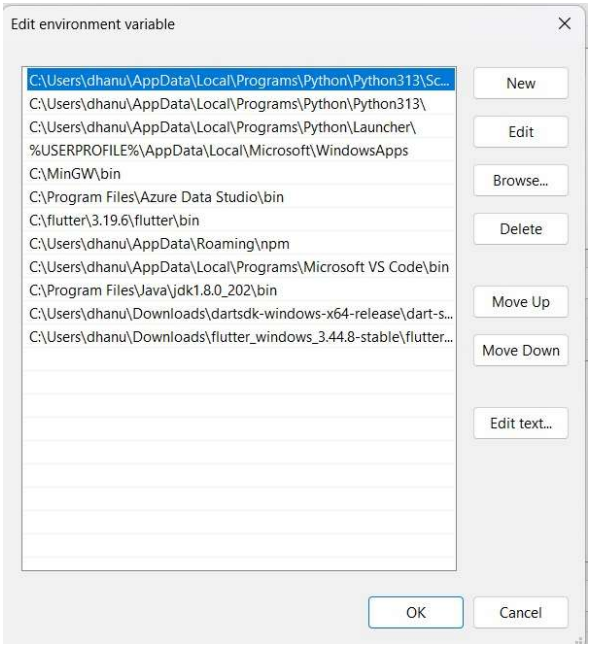


Step 2: Extract Flutter SDK & Dart SDK

- Extract the downloaded ZIP file.
- Move the extracted Flutter folder to a suitable location.

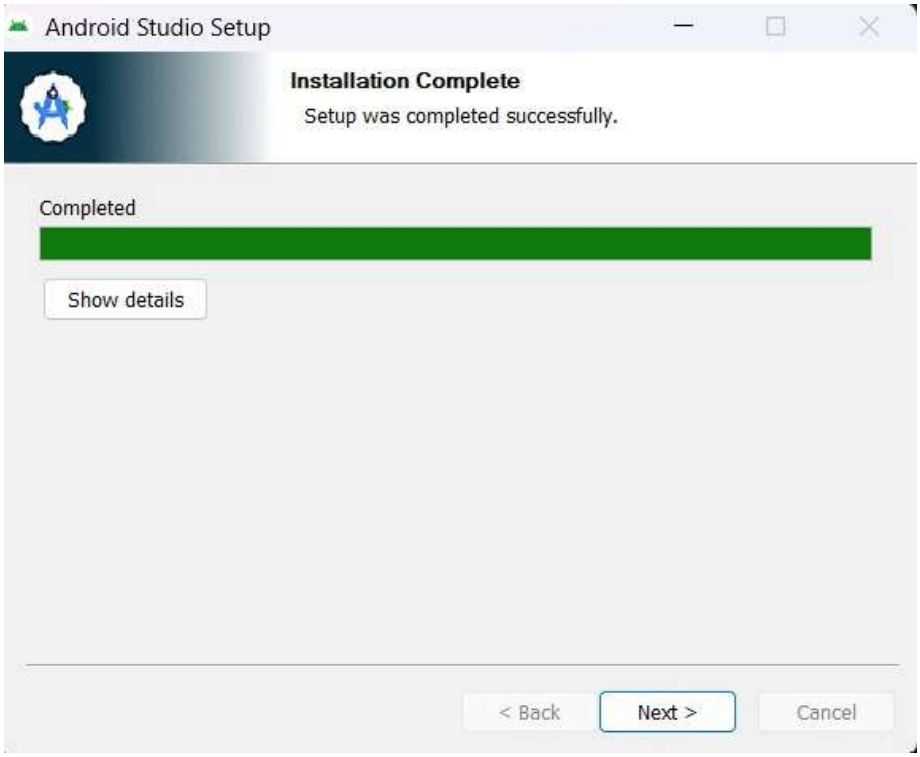
Step 3: Configure Environment Variables

- Open System Properties → Environment Variables.
- Edit the Path variable under System Variables.
- Add the Flutter\bin directory and Dart\bin directory.
- Click OK.



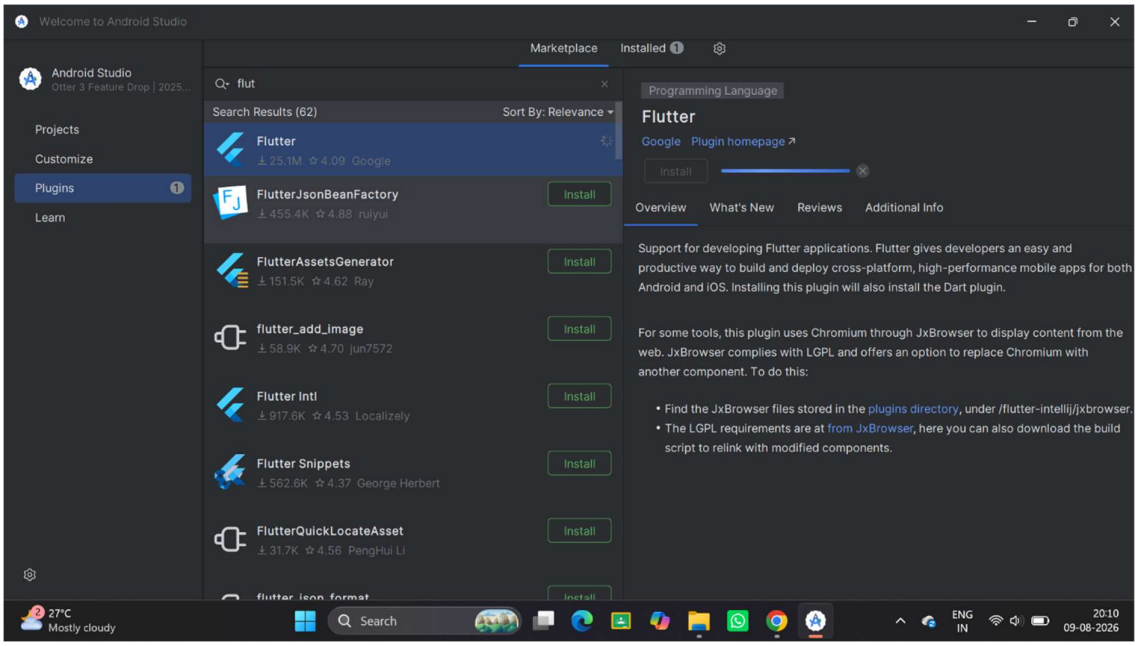
Step 4: Install Android Studio

- Download and install Android Studio.
- During installation, ensure that the following are selected:
- Android SDK
- SDK Platform
- SDK Tools
- AVD

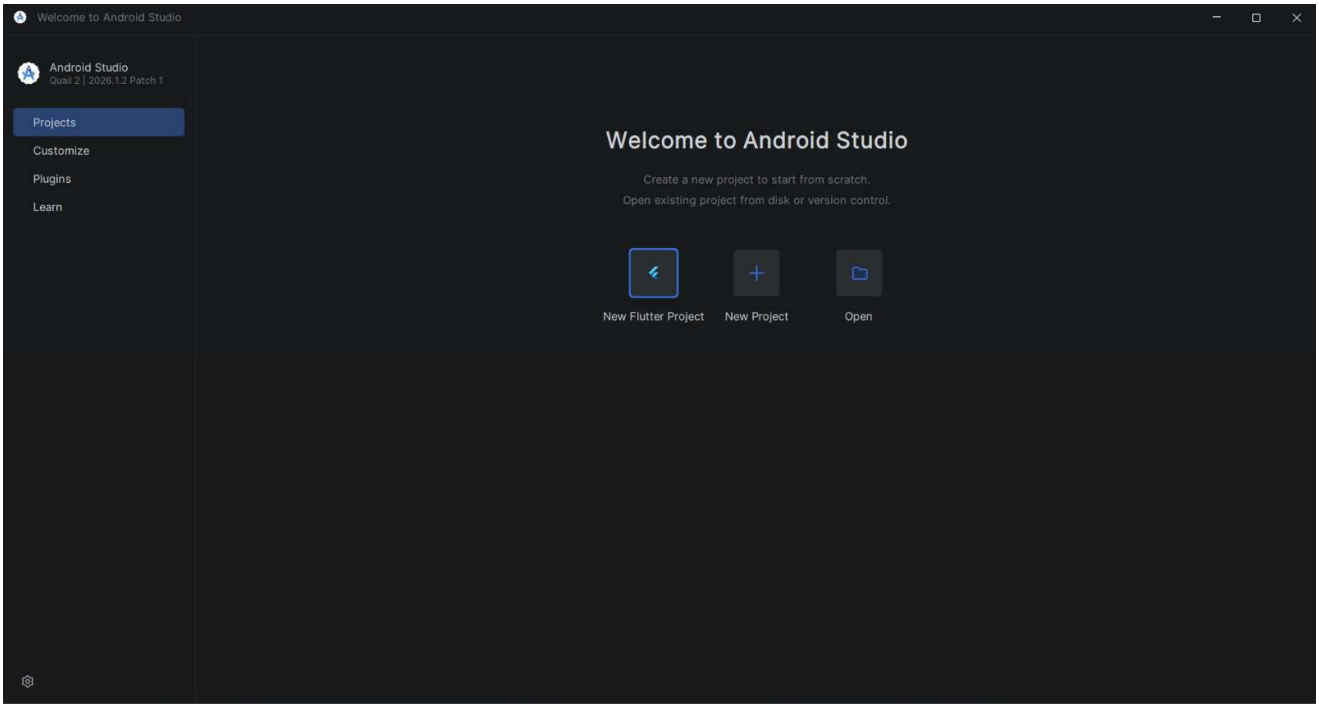


Step 5: Install Flutter, Dart & Dart Plugins

- Open Android Studio.
- Go to Settings → Plugins.
- Search for Flutter and install it.
- The Dart plugin will be installed automatically.
- Restart Android Studio.



Final Output:-



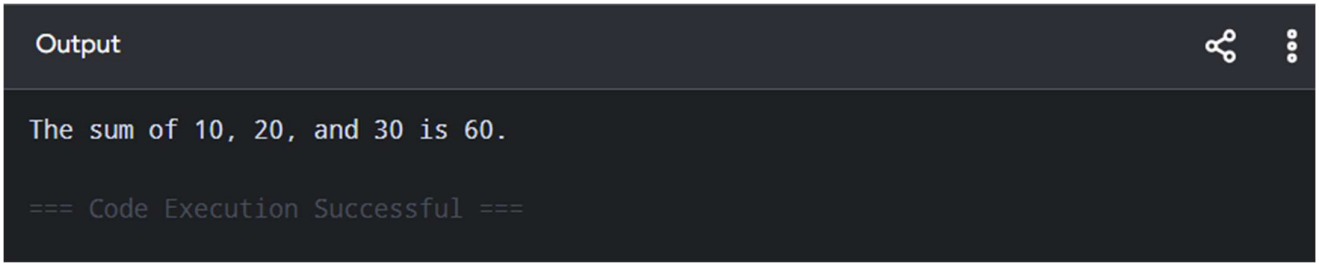
Result:

The Flutter SDK and Dart SDK were successfully installed and configured. The development environment was successfully set up, and a sample Flutter application was created and executed successfully.

1 b) Write a simple Dart program to understand the language basics

```
void main() {
  int num1 = 10;
  int num2 = 20;
  int num3 = 30;
  int sum = num1 + num2 + num3;
  print('The sum of $num1, $num2, and $num3 is $sum.');
}
```

Final Output:-



2a) Explore various Flutter widgets (Text, Image, Container, etc.).

```
import 'package:flutter/material.dart';
void main() {
 runApp(MyApp());
}
class MyApp extends StatefulWidget {
 const MyApp({super.key});
 @override
 State<MyApp> createState() => _MyAppState();
}
class _MyAppState extends State<MyApp> {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
    home: Scaffold(
   appBar: AppBar(
    title: Text('Image'),
   ),
   body: Center(
    child: Container(
     height: 200,
     width: 300,
     child: Image.network('https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcGgRs'),
    ),
   ),
  ));
 }
}
```

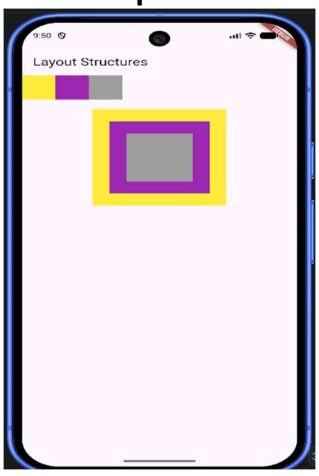
Final Output:-



2 b) Implement different layout structures using Row, Column, and Stack widgets.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text('Combined Layout Example')),
        body: Center(
          child: Column(
            children: <Widget>[
              Row(
                children: <Widget>[
                  Container(color: Colors.yellow, width: 50, height: 50),
                  Container(color: Colors.purple, width: 50, height: 50),
                  Container(color: Colors.grey, width: 50, height: 50),
                ],
              ),
              SizedBox(height: 20),
              Stack(
                alignment: Alignment.center,
                children: <Widget>[
                  Container(color: Colors.yellow, width: 200, height: 200),
                  Container(color: Colors.purple, width: 150, height: 150),
                  Container(color: Colors.grey, width: 100, height: 100),
                ],
              ),
            ],
          ),
        ),
      ),
    );
  }
```

Final Output:-



3 a) Design a responsive UI that adapts to different screen sizes

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return SafeArea(
      child: MaterialApp(
        home: Text('Dhanunjaya'),
      ),
    );
  }
}
```

Final Output:-



3 b) Implement media queries and breakpoints for responsiveness.

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: ResponsiveLayout(),
  );
 }
}

class ResponsiveLayout extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  var mediaQueryData = MediaQuery.of(context);
  var screenWidth = mediaQueryData.size.width;

  if (screenWidth < 600) {
   return Scaffold(
    appBar: AppBar(title: Text('Mobile Layout')),
    body: _buildNarrowContainers(),
   );
  } else if (screenWidth < 1200) {
   return Scaffold(
    appBar: AppBar(title: Text('Ta      blet Layout')),
    body: _buildMediumContainers(),
   );
  } else {
   return Scaffold(
    appBar: AppBar(title: Text('Desktop Layout')),
    body: _buildWideContainers(),
   );
  }
 }

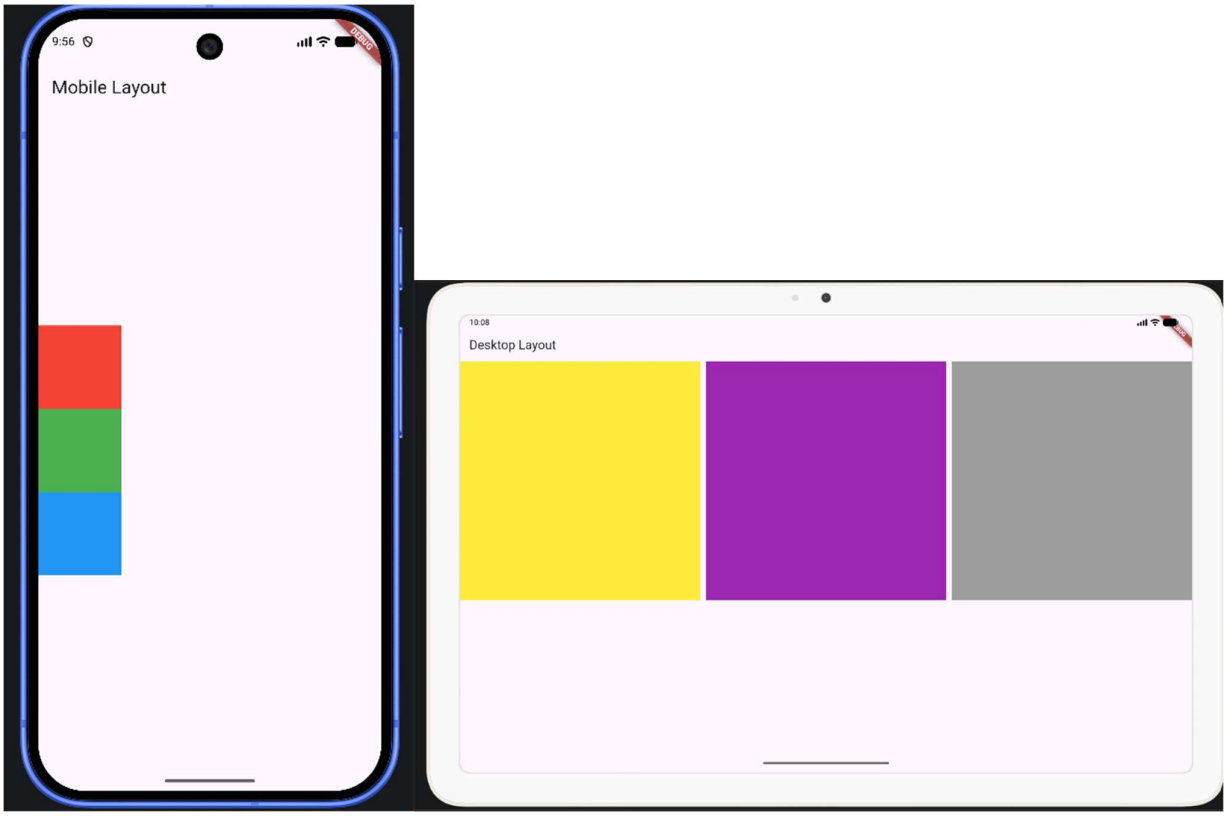
 Widget _buildNarrowContainers() {
  return Column(
   mainAxisAlignment: MainAxisAlignment.center,
   children: <Widget>[
    Container(color: Colors.yellow, width: 100, height: 100),
    Container(color: Colors.purple, width: 100, height: 100),
    Container(color: Colors.grey, width: 100, height: 100),
   ],
  );
 }

 Widget _buildMediumContainers() {
```

```
    return Row(
      mainAxisAlignment: MainAxisAlignment.center,
      children: <Widget>[
        Container(color: Colors.yellow, width: 100, height: 100),
        Container(color: Colors.purple, width: 100, height: 100),
        Container(color: Colors.grey, width: 100, height: 100),
      ],
    );
  }

  Widget _buildWideContainers() {
    return GridView.count(
      crossAxisCount: 3,
      mainAxisSpacing: 10,
      crossAxisSpacing: 10,
      children: <Widget>[
        Container(color: Colors.yellow, width: 100, height: 100),
        Container(color: Colors.purple, width: 100, height: 100),
        Container(color: Colors.grey, width: 100, height: 100),
      ],
    );
  }
}
```

Final Output:-



4 a) Set up navigation between different screens using Navigator.

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}



class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: HomeScreen(),
  );
 }
}
class HomeScreen extends StatelessWidget {
 const HomeScreen({super.key});

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text('Order Page '),
   ),
   body: Center(
    child: ElevatedButton(
     onPressed: () {
      Navigator.push(
       context,
       MaterialPageRoute(builder: (context) => SecondScreen()),
      );
     },
     child: Text('Go to Cart'),
    ),
   ),
  );
 }
}
class SecondScreen extends StatelessWidget {
 const SecondScreen({super.key});

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
```

```
      title: Text('Cart page'),
    ),
    body: Center(
      child: ElevatedButton(
        onPressed: () {
          Navigator.pop(context);
        },
        child: Text('Go back to Order Page'),
      ),
    ),
  );
 }
}
```

Final Output:-



4 b) Implement navigation with named routes.

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      initialRoute: '/',
      routes: {
        '/': (context) => HomeScreen(),
        '/second': (context) => SecondScreen(),
      },
    );
  }
}


class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

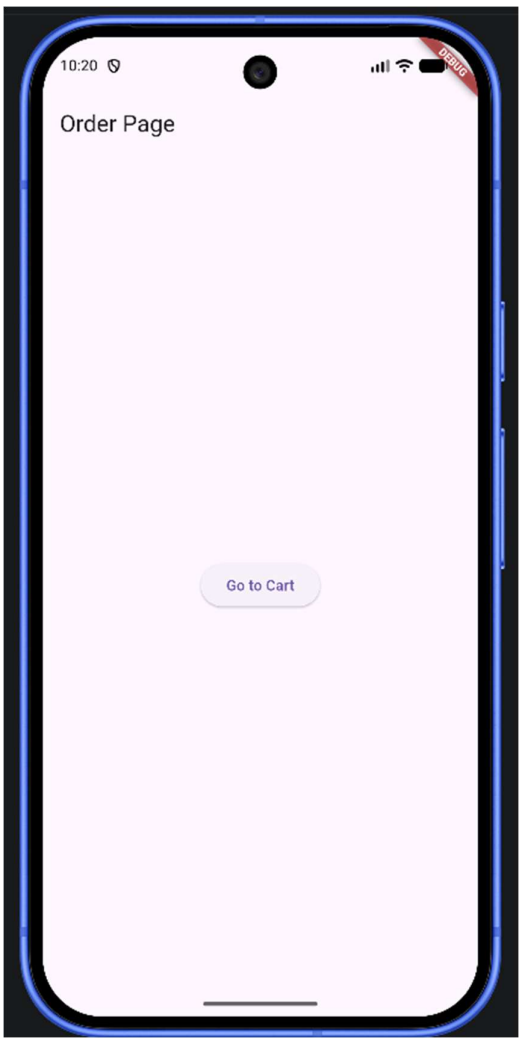
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Order Page'),
      ),
      body: Center(
        child: ElevatedButton(
          onPressed: () {
            Navigator.pushNamed(context, '/second');
          },
          child: Text('Go to Cart'),
        ),
      ),
    );
  }
}

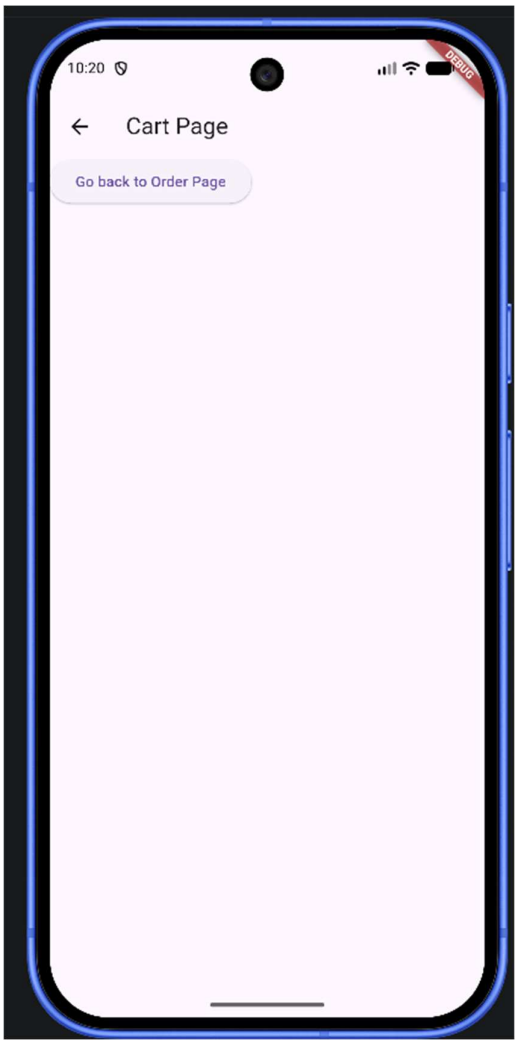
class SecondScreen extends StatelessWidget {
  const SecondScreen({super.key});

  @override
```

```
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Cart Page'),
      ),
      body: ElevatedButton(
        onPressed: () {
          Navigator.pop(context);
        },
        child: Text('Go back to Order Page'),
      ),
    );
  }
}
```

Final Output:-

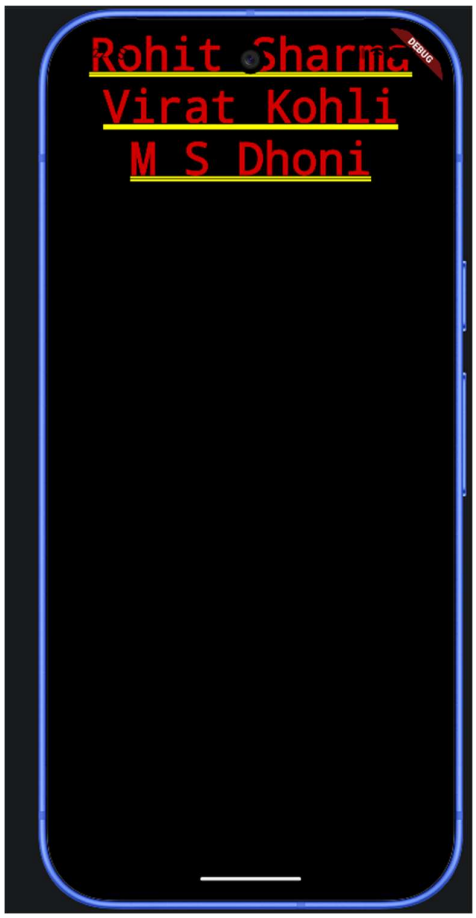




5 a) Learn about stateful and stateless widgets

```
import 'package:flutter/material.dart';
void main() {
 runApp(MyApp());
}
class MyApp extends StatefulWidget {
 const MyApp({super.key});
 @override
 State<MyApp> createState() => _MyAppState();
}
class _MyAppState extends State<MyApp> {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: Column(
    children: [Text('Rohit Sharma'), Text('Virat Kohli'), Text('M S Dhoni')],
   ),
  );
}
}
```

Final Output:-



5 (A) (2)

```
import 'package:flutter/material.dart';
void main() {
 runApp(MyApp());
}
class MyApp extends StatelessWidget {
 const MyApp({super.key});
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: Text('Hello'),
  );
 }
}
```

Final Output:-

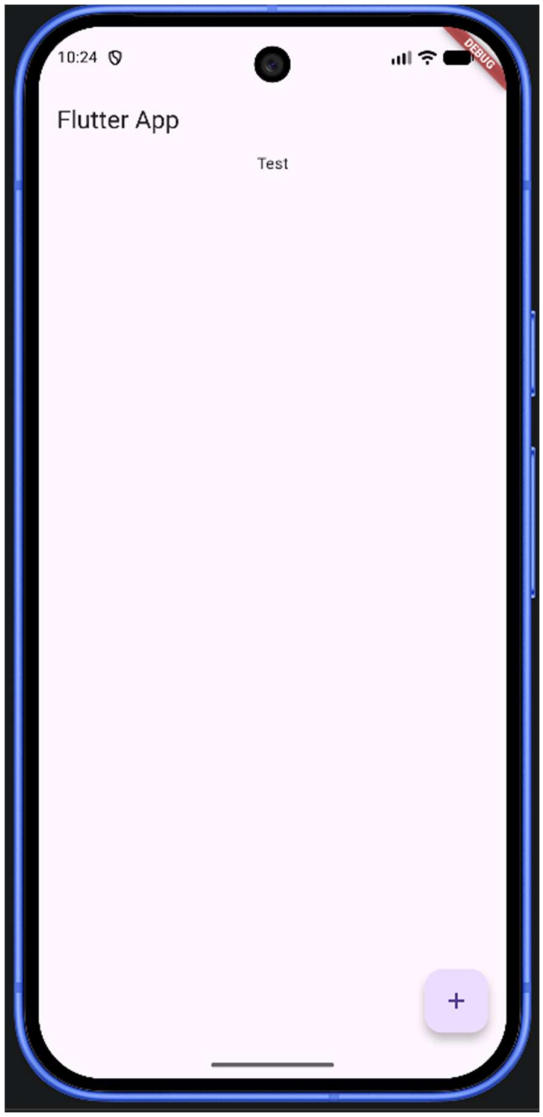


5 b) Implement state management using set State and Provider.

```
import 'package:flutter/material.dart';
void main() {
 runApp(MyApp());
}
class MyApp extends StatefulWidget {
 const MyApp({super.key});
 @override
 State<MyApp> createState() => _MyAppState();
}
class _MyAppState extends State<MyApp> {
 String value = "Test"; //variable initialized
  void clickMe() {
   setState(() {
     value = "Test done";
   });
 }
 @override
 Widget build(BuildContext context) {

   return MaterialApp(
home: Scaffold(
      appBar: AppBar(
        title: Text('Flutter app'),
      ),
      body: Center(
        child: Column(
          children: [
            Text('$value'),
          ],
        ),
      ),
      floatingActionButton: FloatingActionButton(
        child: Icon(Icons.add),
        onPressed: clickMe,
      ),
    ),
   );
 }
}
```

Final Output:-



6 a) Create custom widgets for specific UI elements.

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 const MyApp({super.key});
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: HomeScreen(),
  );
 }
}

//Home screen - class
class HomeScreen extends StatelessWidget {
 const HomeScreen({super.key});

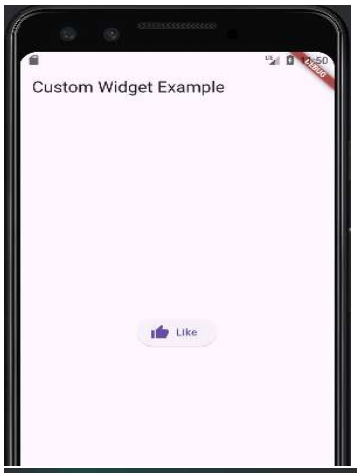
 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text("Custom Widget Example"),
   ),
   body: Center(
    child: IconTextButton(
      icon: Icons.thumb_up, //variable 1
      text: 'Like', //variable 2
      onPressed: () {
       //variable 3
       print('Button Pressed!');
      }),
   ),
  );
 }
}

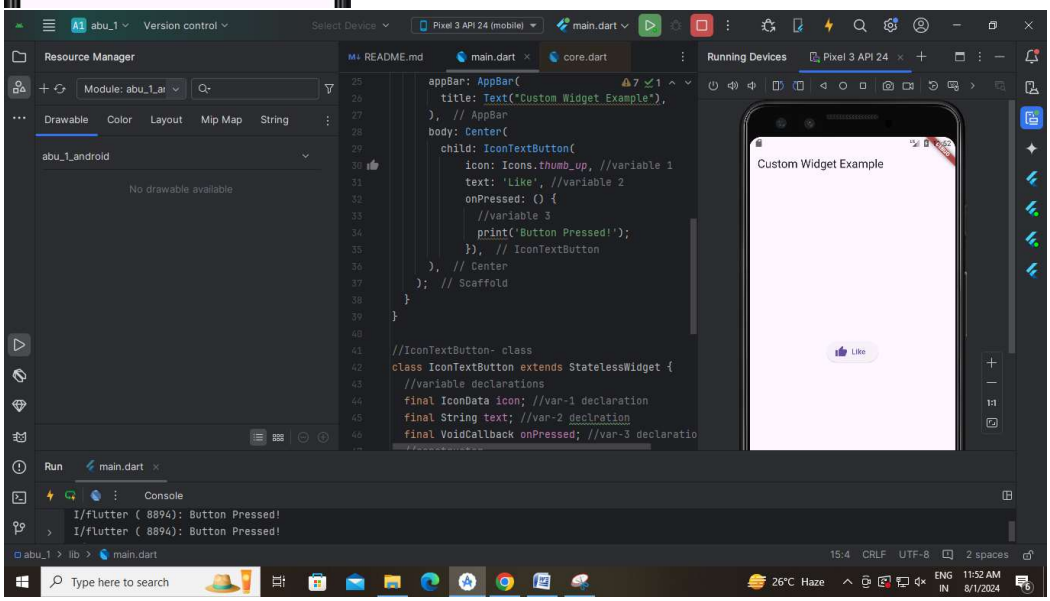
//IconTextButton- class
class IconTextButton extends StatelessWidget {
 //variable declarations
 final IconData icon; //var-1 declaration
 final String text; //var-2 declration
 final VoidCallback onPressed; //var-3 declaration
 //constructor
 IconTextButton({
```

```
    required this.icon,
    required this.text,
    required this.onPressed,
  });

  @override
  Widget build(BuildContext context) {
    return ElevatedButton.icon(
      onPressed: onPressed,
      icon: Icon(icon),
      label: Text(text),
    );
  }
}
```

Final Output:-





6 b) Apply styling using themes and custom styles.

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}
class MyApp extends StatelessWidget {
 const MyApp({super.key});
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: HomeScreen(),
  );
 }
}

//Home screen - class
class HomeScreen extends StatelessWidget {
 const HomeScreen({super.key});

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text("Custom Widget Example"),
   ),
   body: Center(
    child: IconTextButton(
      icon: Icons.thumb_up, //variable 1
      text: 'Like', //variable 2
      onPressed: () {
       //variable 3
       print('Button Pressed!');
      }),
   ),
  );
 }
}

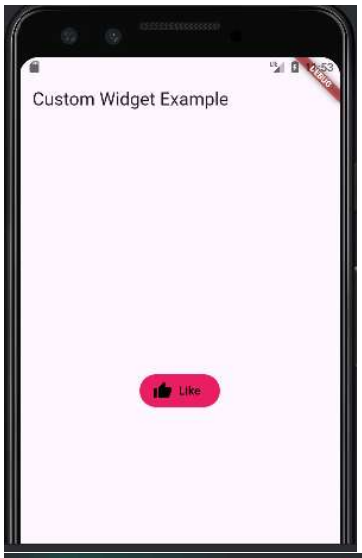
//IconTextButton- class
class IconTextButton extends StatelessWidget {
 //variable declarations
 final IconData icon; //var-1 declaration
 final String text; //var-2 declration
 final VoidCallback onPressed; //var-3 declaration
 //constructor
 IconTextButton({
  required this.icon,
```

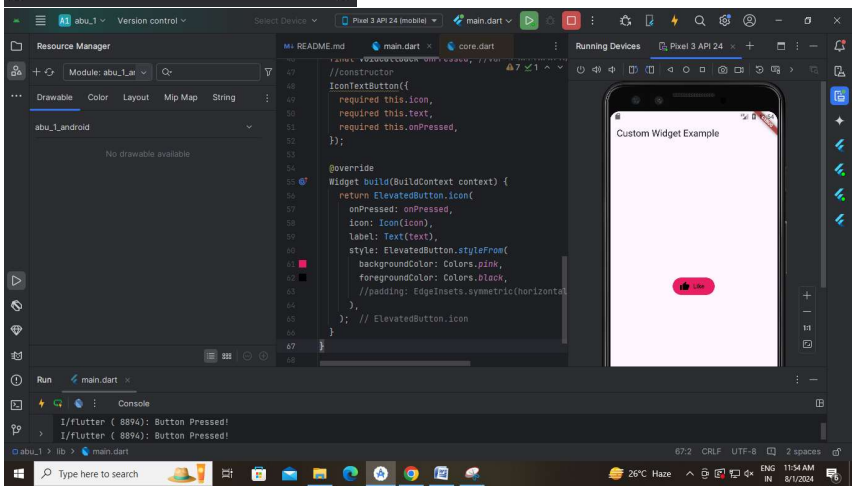
```
    required this.text,
    required this.onPressed,
  });

  @override
Widget build(BuildContext context) {
    return ElevatedButton.icon(
      onPressed: onPressed,
      icon: Icon(icon),
      label: Text(text),
      style: ElevatedButton.styleFrom(
        backgroundColor: Colors.pink,
        foregroundColor: Colors.black,

      ),
    );
  }
}
```

Final Output:-





7 a) Design a form with various input ields.

```
import 'package:flutter/material.dart';
void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: Scaffold(
    appBar: AppBar(
     title: Text('Simple Form Example'),
    ),
    body: Padding(
     padding: const EdgeInsets.all(16.0),
     child: SimpleForm(),
    ),
   ),
  );
 }
}

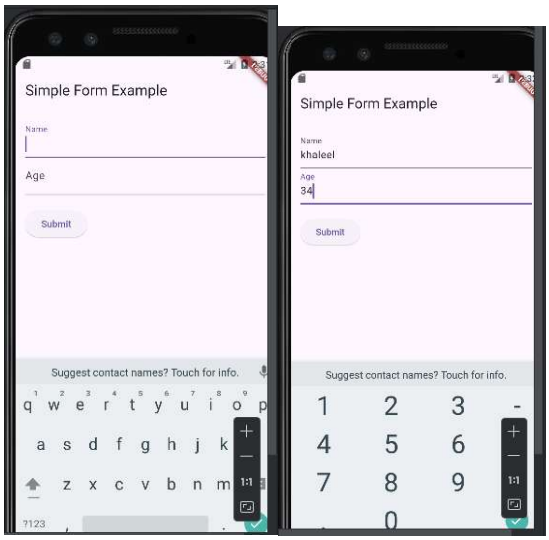
class SimpleForm extends StatefulWidget {
 @override
 _SimpleFormState createState() => _SimpleFormState();
}

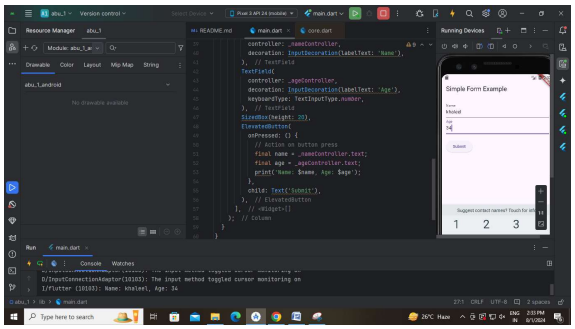
class _SimpleFormState extends State<SimpleForm> {
 final _nameController = TextEditingController();
 final _ageController = TextEditingController();

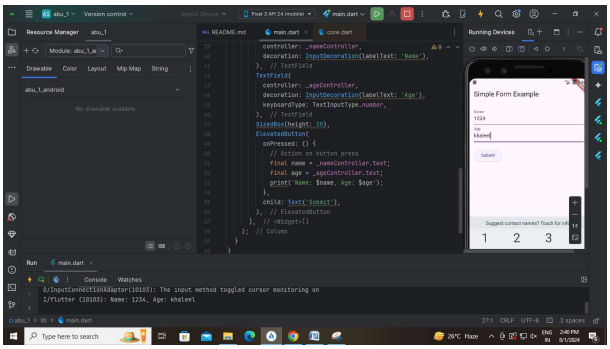
 @override
 Widget build(BuildContext context) {
  return Column(
   crossAxisAlignment: CrossAxisAlignment.start,
   children: <Widget>[
    TextField(
     controller: _nameController,
     decoration: InputDecoration(labelText: 'Name'),
    ),
    TextField(
     controller: _ageController,
     decoration: InputDecoration(labelText: 'Age'),
     keyboardType: TextInputType.number,
    ),
```

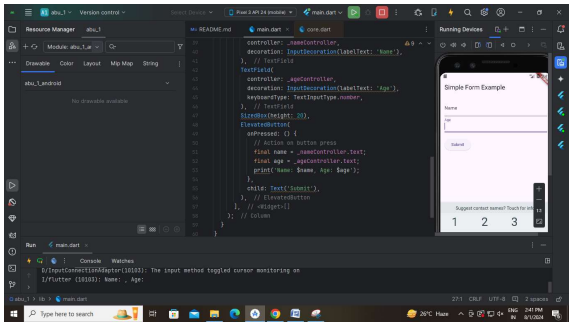
```
      SizedBox(height: 20),
      ElevatedButton(
       onPressed: () {
         // Action on button press
         final name = _nameController.text;
         final age = _ageController.text;
         print('Name: $name, Age: $age');
       },
       child: Text('Submit'),
      ),
    ],
  );
}
```

Final Output:-









7 b) Implement form validation and error handling.

```
import 'package:flutter/material.dart';

void main() (
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: Scaffold(
    appBar: AppBar(
     title: Text('Simple Form with Validation'),
    ),
    body: Padding(
     padding: const EdgeInsets.all(16.0),
     child: SimpleForm(),
    ),
   ),
  );
 }
}

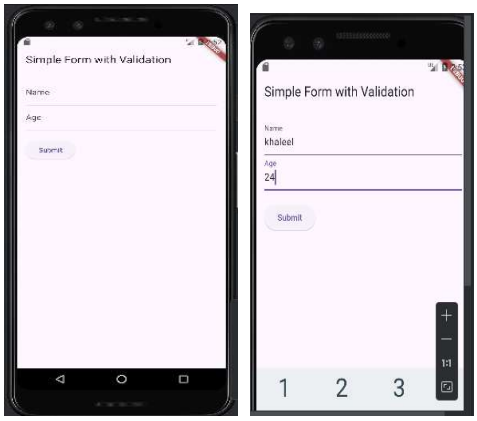
class SimpleForm extends StatefulWidget {
 @override
 _SimpleFormState createState() => _SimpleFormState();
}

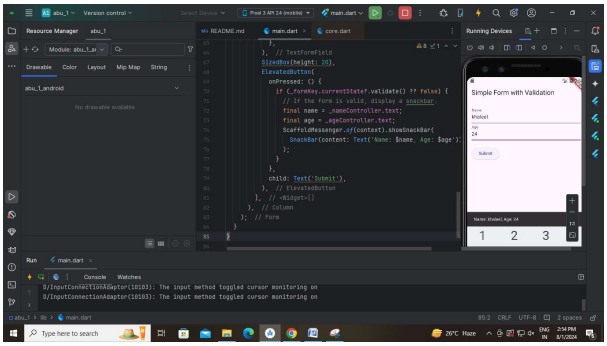
class _SimpleFormState extends State<SimpleForm> {
 final _formKey = GlobalKey<FormState>();

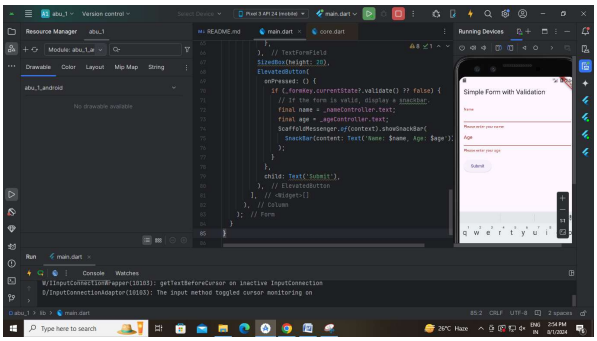
 final _nameController = TextEditingController();
 final _ageController = TextEditingController();
 @override
 Widget build(BuildContext context) {
  return Form(
   key: _formKey,
   child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: <Widget>[
     TextFormField(
      controller: _nameController,
      decoration: InputDecoration(labelText: 'Name'),
      validator: (value) {
       if (value == null || value.isEmpty) {
        return 'Please enter your name';
       }
```

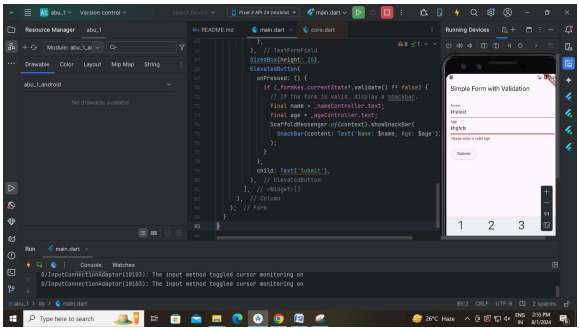
```
            return null;
          },
        ),
        TextFormField(
          controller: _ageController,
          decoration: InputDecoration(labelText: 'Age'),
          keyboardType: TextInputType.number,
          validator: (value) {
            if (value == null || value.isEmpty) {
              return 'Please enter your age';
            }
            final age = int.tryParse(value);
            if (age == null || age <= 0) {
              return 'Please enter a valid age';
            }
            return null;
          },
        ),
        SizedBox(height: 20),
        ElevatedButton(
          onPressed: () {
            if (_formKey.currentState?.validate() ?? false) {
              // If the form is valid, display a snackbar.
              final name = _nameController.text;
              final age = _ageController.text;
              ScaffoldMessenger.of(context).showSnackBar(
                SnackBar(content: Text('Name: $name, Age: $age')),
              );
            }
          },
          child: Text('Submit'),
        ),
      ],
    ),
  );
 }
}
```

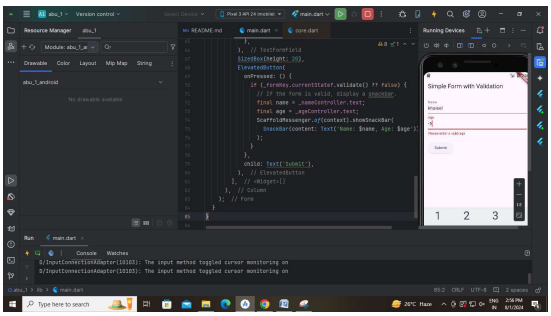
Final Output:-

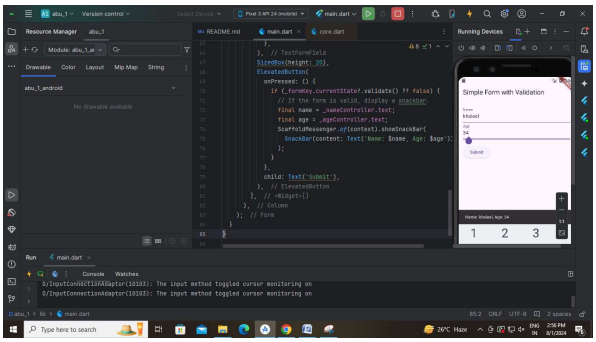












8 a) Add animations to UI elements using Flutter's animation framework.

```
import 'package:flutter/material.dart';

void main() {
          runApp(MyApp());
}
class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: HomeScreen(),
  );
 }
}

class HomeScreen extends StatelessWidget {
 const HomeScreen({super.key});

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text("Custon widget with animation"),
   ),
   body: Center(
    child: AnimatedIconTextButton(
       icon: Icons.thumb_up,
       text: 'Like',
       onPressed: () {
        print("Button Pressed!");
       }),
   ),
  );
 }
}

class AnimatedIconTextButton extends StatefulWidget {
 final IconData icon;
 final String text
 final VoidCallback onPressed;
 //constructors
 const AnimatedIconTextButton({
  Key? key,
  required this.icon,
  required this.text,
```

```
    required this.onPressed,
  }) : super(key: key);

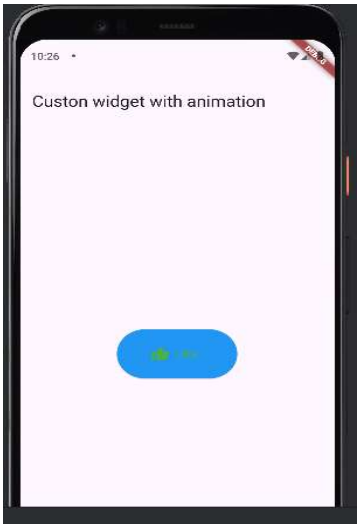
  @override
  State<AnimatedIconTextButton> createState() => _AnimatedIconTextButtonState();
}


class _AnimatedIconTextButtonState extends State<AnimatedIconTextButton> {

  bool _isPressed = false;
  void _handlePress() {
    setState(() {
      _isPressed = !_isPressed; //on->off//off->on
    });
    widget.onPressed();
  }

  @override
  Widget build(BuildContext context) {
    return AnimatedContainer(
      duration: Duration(milliseconds: 300),
      width: _isPressed ? 200 : 150,
      height: _isPressed ? 100 : 70,
      child: ElevatedButton.icon(
        onPressed: _handlePress,
        icon: Icon(widget.icon),
        label: Text(widget.text),
        style: ElevatedButton.styleFrom(
          backgroundColor: Colors.blue,
          foregroundColor: Colors.green,
        ),
      ),
    );  } }
```

Final Output:-



8 b) Experiment with different types of animations (fade, slide)

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   home: HomeScreen(),
  );
 }
}

class HomeScreen extends StatelessWidget {
 const HomeScreen({super.key});

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text("Custon widget with animation"),
   ),
   body: Center(
    child: FadeIconTextButton(
      icon: Icons.thumb_up,
      text: 'Like',
      onPressed: () {
       print("Button Pressed!");
      }),
   ),
  );
 }
}

class FadeIconTextButton extends StatefulWidget {
 final IconData icon;
 final String text;
 final VoidCallback onPressed;
 //constructors
 const FadeIconTextButton({
  Key? key,
  required this.icon,
  required this.text,
```

```
    required this.onPressed,
  }) : super(key: key);

  @override
  State<FadeIconTextButton> createState() => _FadeIconTextButtonState();
}
class _FadeIconTextButtonState extends State<FadeIconTextButton>
    with SingleTickerProviderStateMixin {
  late AnimationController _controller;
  late Animation<double> _animation;
  @override
  void initState() {
    super.initState();
    _controller = AnimationController(
      duration: const Duration(milliseconds: 500),
      vsync: this,
    );
    _animation = CurvedAnimation(
      parent: _controller,
      curve: Curves.easeInOut,
    );
  }

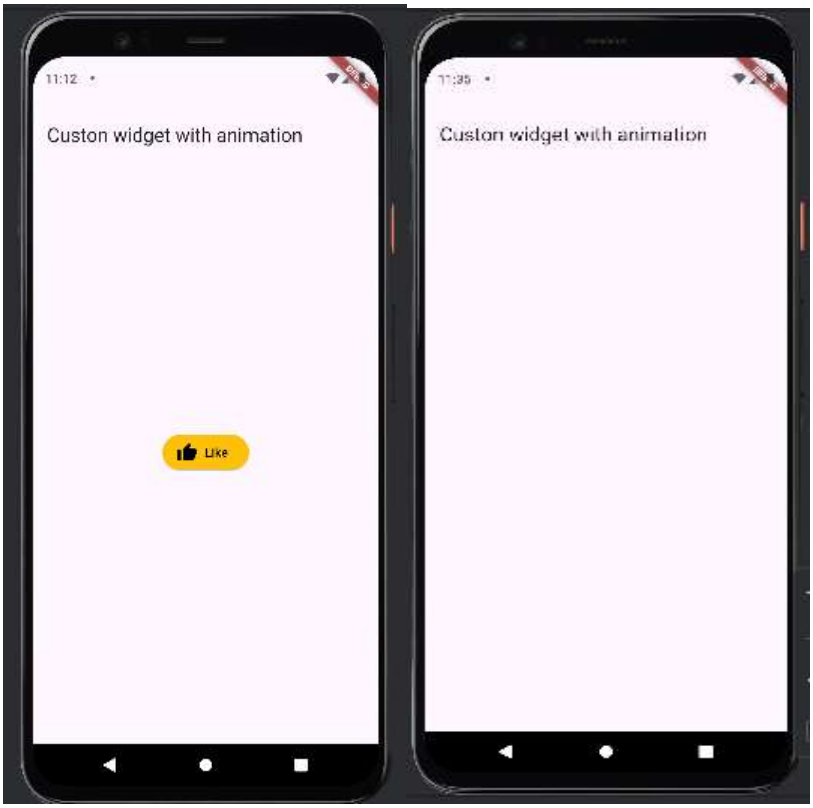
  void _handlePress() {
    if (_controller.isCompleted) {
      _controller.reverse();
    } else {
      _controller.forward();
    }
    widget.onPressed();
  }

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return FadeTransition(
      opacity: _animation,
      child: ElevatedButton.icon(
        onPressed: _handlePress,
        icon: Icon(widget.icon),
        label: Text(widget.text),
        style: ElevatedButton.styleFrom(
          backgroundColor: Colors.amber,
```

```
        foregroundColor: Colors.black,
      ),
    ),
  );
 }
}
```

Final Output:-

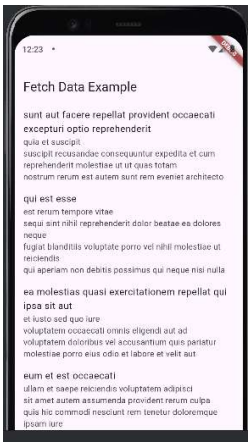


9 a) Fetch data from a REST API.
b) Display the fetched data in a meaningful way in the UI.

```
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
import 'dart:convert';
void main() {
 runApp(MyApp());
}
class MyApp extends StatelessWidget {
 Widget build(BuildContext context) {
  return MaterialApp(
   home: HomeScreen(),
  );
 }
}
class HomeScreen extends StatefulWidget {
 @override
 _HomeScreenState createState() => _HomeScreenState();
}
class _HomeScreenState extends State<HomeScreen> {
 late Future<List<Post>> _futurePosts;
 @override
 void initState() {
  super.initState();
  _futurePosts = fetchPosts();
 }
 Future<List<Post>> fetchPosts() async {
  final response =
     await http.get(Uri.parse('https://jsonplaceholder.typicode.com/posts'));
  if (response.statusCode == 200) {
   List jsonResponse = json.decode(response.body);
   return jsonResponse.map((post) => Post.fromJson(post)).toList();
  } else {
   throw Exception('Failed to load posts');
  }
 }
 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text('Fetch Data Example'),
   ),
   body: Center(
    child: FutureBuilder<List<Post>>(
     future: _futurePosts,
     builder: (context, snapshot) {
```

```
          if (snapshot.connectionState == ConnectionState.waiting) {
            return CircularProgressIndicator();
          } else if (snapshot.hasError) {
            return Text('${snapshot.error}');
          } else if (snapshot.hasData) {
            List<Post>? posts = snapshot.data;
            return ListView.builder(
              itemCount: posts!.length,
              itemBuilder: (context, index) {
                return ListTile(
                  title: Text(posts[index].title),
                  subtitle: Text(posts[index].body),
                );
              },
            );
          } else {
            return Text('No data found');
          }
        },
      ),
    ),
  );
 }
}
class Post {
 final int id;
 final String title;
 final String body;
 Post({required this.id, required this.title, required this.body});
 factory Post.fromJson(Map<String, dynamic> json) {
  return Post(
    id: json['id'],
   title: json['title'],
   body: json['body'],
 );
 }}
```

Final Output:-



10 a) Write unit tests for UI components.

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   title: 'Flutter Demo',
   theme: ThemeData(
    primarySwatch: Colors.blue,
   ),
   home: HomeScreen(),
  );
 }
}

class HomeScreen extends StatefulWidget {
 @override
 _HomeScreenState createState() => _HomeScreenState();
}

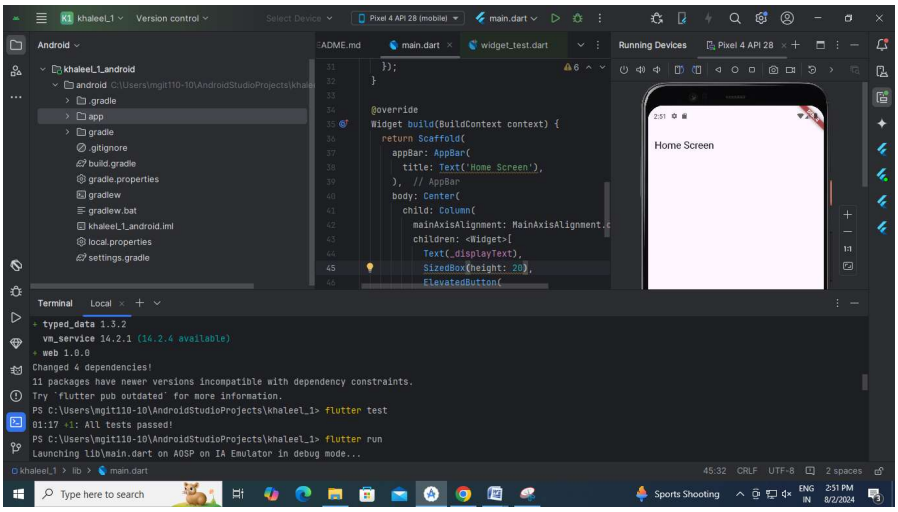
class _HomeScreenState extends State<HomeScreen> {
 String _displayText = 'Hello';

 void _updateText() {
  setState(() {
   _displayText = 'Button Pressed';
  });
 }

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text('Home Screen'),
   ),
   body: Center(
    child: Column(
     mainAxisAlignment: MainAxisAlignment.center,
     children: <Widget>[
      Text(_displayText),
      SizedBox(height: 20),
      ElevatedButton(
```

```
            onPressed: _updateText,
            child: Text('Press Me'),
          ),
        ],
      ),
    ),
  );
 }
}
import 'package:flutter/material.dart';
import 'package:flutter_test/flutter_test.dart';
import 'package:untitled/main.dart'; //url might change
void main() {
 testWidgets('text changes', (WidgetTester tester) async {
  //build the app
  await tester.pumpWidget(MyApp());
  //test-1 verifying the initial text'Hello
  expect(find.text("Hello"), findsOneWidget);
  expect(find.text("Button Pressed"), findsNothing);
  //test-2 to tap the button
  await tester.tap(find.byType(ElevatedButton));
  await tester.pump();
  //test-3 verify the text
  expect(find.text("Hello"), findsNothing);
  expect(find.text("Button Pressed"), findsOneWidget);
 });
}
```

Final Output:-



10 b) Use Flutter's debugging tools to identify and fix issues.

```
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   title: 'Flutter Demo',
   theme: ThemeData(
    primarySwatch: Colors.blue,
   ),
   home: HomeScreen(),
  );
 }
}

class HomeScreen extends StatefulWidget {
 @override
 _HomeScreenState createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
 String _displayText; // Variable declared but not initialized

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text('Home Screen'),
   ),
   body: Center(
    child: Column(
     mainAxisAlignment: MainAxisAlignment.center,
     children: <Widget>[
      Text(_displayText), // Error: _displayText is used before being initialized
      SizedBox(height: 20),
      ElevatedButton(
       onPressed: () {
        setState(() {
         _displayText = 'Button Pressed';
        });
       },
```

```
            child: Text('Press Me'),
          ),
        ],
      ),
    ),
  );
 }
}
//corrected code
import 'package:flutter/material.dart';

void main() {
 runApp(MyApp());
}

class MyApp extends StatelessWidget {
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   title: 'Flutter Demo',
   theme: ThemeData(
    primarySwatch: Colors.blue,
   ),
   home: HomeScreen(),
  );
 }
}

class HomeScreen extends StatefulWidget {
 @override
 _HomeScreenState createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
 String _displayText=''; // Variable declared and initialized

 @override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(
    title: Text('Home Screen'),
   ),
   body: Center(
    child: Column(
     mainAxisAlignment: MainAxisAlignment.center,
     children: <Widget>[
      Text(_displayText), // Error: _displayText is used before being initialized
      SizedBox(height: 20),
```

```
        ElevatedButton(
          onPressed: () {
            setState(() {
              _displayText = 'Button Pressed';
            });
          },
          child: Text('Press Me'),
        ),
      ],
    ),
  ),
 );
 }
}
```

Final Output:-

